



Water We Doing Here?

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Fields of Interest: Environmental Science, Ecology

Brief Overview:

Mentees will get to know BEAM through a lesson about the environment. Mentees will learn about ecosystems, the water cycle, and the carbon cycle by making their own terrariums.

Agenda:

- Introduction (5 min)
- Module 1: Well-come to BEAM! (10-15 min)
- Module 2: It's raining tacos! (15-20 min)
- Module 3: BYOB (Bring your Own Bugs) (15-20 min)
- Conclusion (5 min)

Main Teaching Goals/Key Terms:

- Ecosystem
- Biotic and Abiotic Factors
- Water Cycle
- Clouds
- Carbon Cycle

Background for Mentors

Module 2

- Water Cycle
- Evaporation
- Condensation
- Clouds
- Precipitation
- Transpiration

The **water cycle** is a biogeochemical process involving the continuous change of water on, above, and below Earth's surface. The amount of water in all forms has been mostly constant over time, but the proportions of water in each *form* has changed depending on the climate.

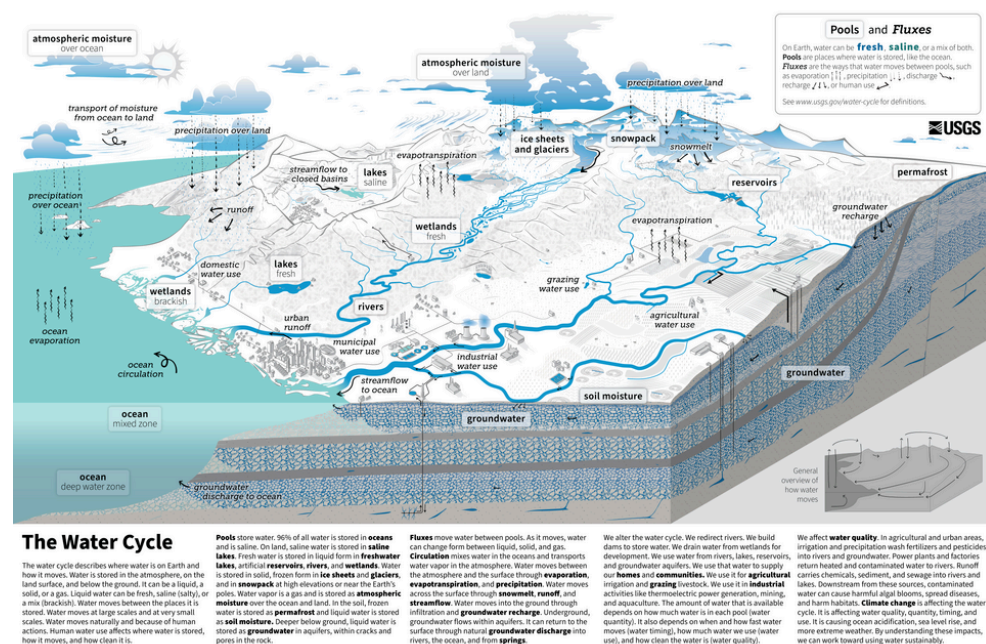


Figure 1: Super complicated water cycle diagram

One component of the **water cycle** is that water first **evaporates**, meaning it transforms from liquid to gas, from bodies of water, including puddles, creeks, rivers, lakes, oceans, etc. The water is then held in the air which we experience as humidity. This water vapor rises into the sky where it then cools down and **condenses**, turning from gas back to water, and if it's cold enough, into ice and snow. These then collect with dust to form **clouds**.

When droplets or snow grow too large, they are too heavy to stay in the clouds and so they fall down; this is **precipitation**.

Transpiration is the flow of water through a plant, starting through its roots to evaporating at the leaves or flowers of a plant. Like sweating in humans, this allows plants to passively cool!

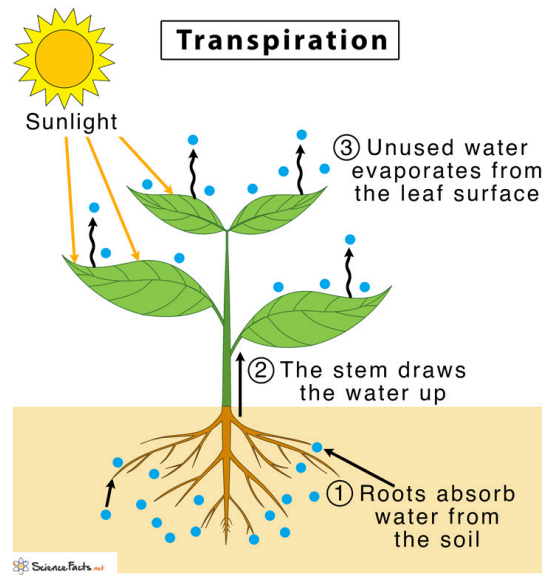


Figure 2: Transpiration diagram

Module 3

- Ecosystem
- Biotic Factors
- Abiotic Factors
- Carbon
- Carbon Cycle
- Photosynthesis
- Respiration
- Decomposition

Ecosystems are communities of **biotic factors** interacting with the **abiotic factors** in their environment.

Biotic factors are living or once-living things, including organisms like animals, plants, bacteria, as well as decomposing animal and plant matter. The living things in an ecosystem interact with and depend on one another for survival.

Abiotic factors are the non-living components of an ecosystem. This includes resources such as sunlight, water, soil, and minerals, as well as characteristics like temperature, climate, and elevation.

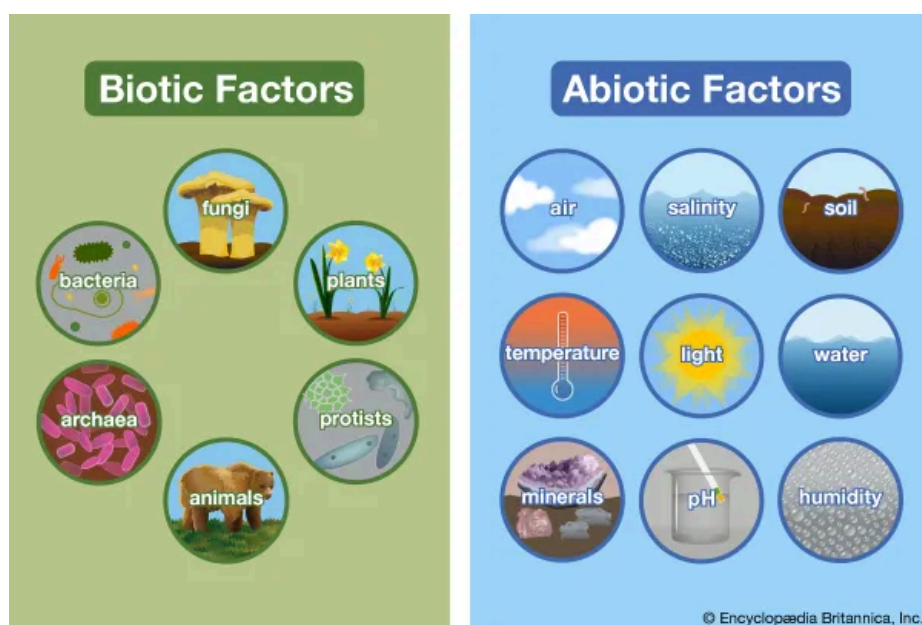


Figure 1: Examples of biotic and abiotic factors.

Living things on Earth are carbon-based, meaning they are mainly made of the element **carbon**. Much like water, carbon cycles predictably between different sources, mainly in the form of carbon dioxide. The carbon cycle involves both biotic and abiotic factors.

Plants and algae perform **photosynthesis** to make energy. This process uses carbon dioxide and expels oxygen as a byproduct. Plants get the carbon dioxide they need from **respiration**, which is the process of animals breaking down food into energy their bodies can use. Respiration uses the oxygen that plants make and expels carbon dioxide as a byproduct. Carbon dioxide is released back into the surrounding environment (soil, atmosphere, etc) by

decomposition, which is the process of fungi and bacteria breaking down dead plant and animal matter into its carbon and other constituents.

While the carbon cycle naturally maintains balance on its own, some human activities release excess carbon dioxide back into the environment at unsustainable rates. The combustion of fossil fuels puts pressure on the ocean and the atmosphere to hold more carbon than they should, resulting in negative effects like global warming and ocean acidification.

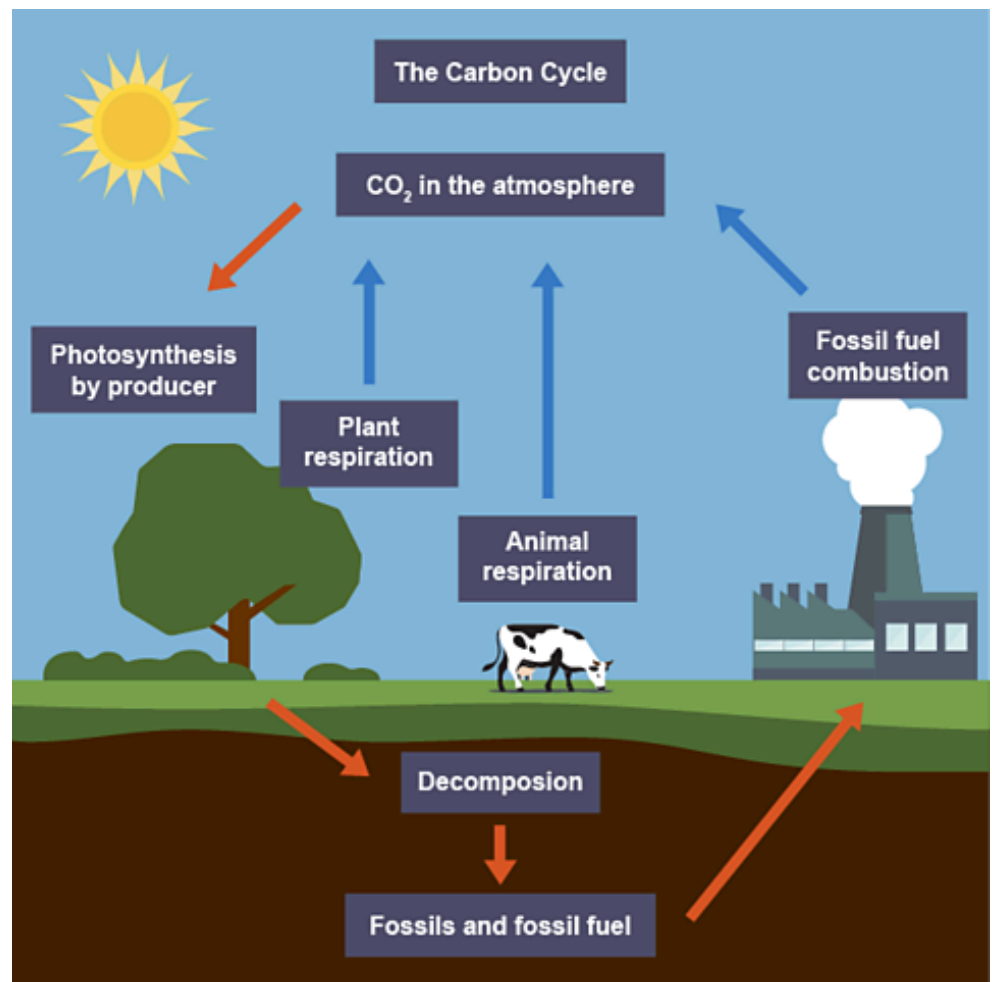


Figure 2: The Carbon Cycle. Arrows represent carbon flow.

Ecosystems are mainly characterized by differences in their abiotic factors. The types of organisms that are able to live in a particular ecosystem vary by temperature, elevation, and the availability of sunlight, water, and nutrients.

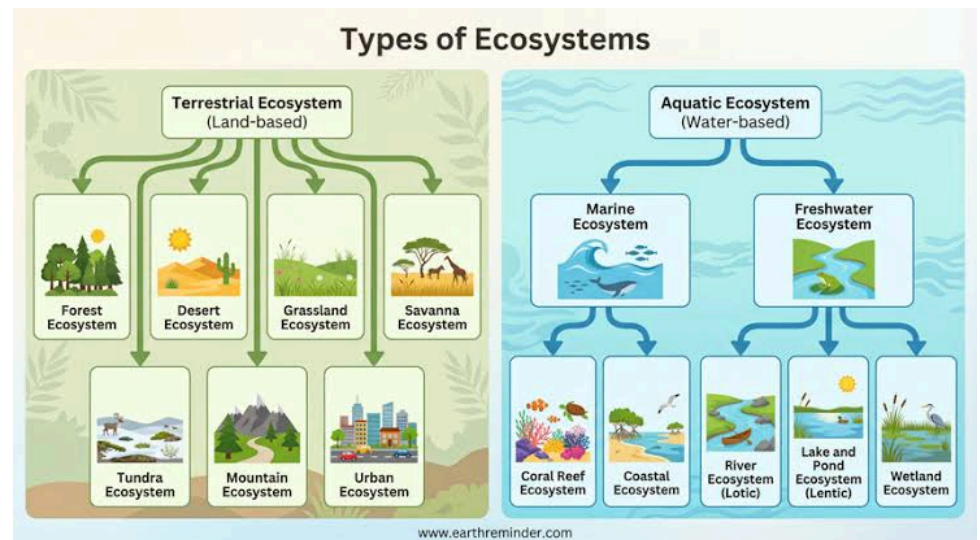


Figure 3: Ecosystems are classified based on abiotic factors. How do these factors vary between types of ecosystems?

We are making terrariums to model ecosystems. Our biotic factors are chia seeds, which will produce oxygen and pull carbon dioxide. Our abiotic factors are soil, water, pebbles, and charcoal. Soil is the medium for the nutrients our chia seeds need to grow, and water is a vital nutrient for photosynthesis. Pebbles provide space away from the roots to store water to prevent water logging and root rot. The activated charcoal absorbs toxins and waste products that harmful bacteria or fungi need to survive. Our terrariums are able to stay fully sealed because the nutrients inside, like water and carbon, cycle throughout it in closed loops.

Introduction

Concepts to Introduce • Ecosystems are collections of living things that work together with non-living things to survive and thrive. • Nutrients like water and carbon cycle in predictable patterns throughout ecosystems. • A terrarium is a miniature model of an ecosystem that showcases how nutrient cycles work in closed loops.	Questions to Pique Interest • Who do you think can become a scientist? • What do you think scientists do? What jobs/careers do they have? • How long do you think you can live without water? • What are living things made out of? • Is dirt dead or alive?
Scientists, Current and Past Events • Greta Thunberg - Swedish activist who works towards addressing climate change and social issues. • Some human activities can make water and carbon cycles fall out of balance. Sustainability initiatives aim to regulate how humans use water and carbon. • Rapid deforestation - Cutting down trees removes vegetation that returns water vapor back into the air, which can change rainfall patterns ◦ Also contributes to the loss of biodiversity in environments	Careers and Applications • Environmental scientists study the natural world and how it changes with time, and how human interactions with it influence the environment and human health. • Hydraulic engineers work on the flow of fluids, particularly water and sewage. • Environmental policy makers work towards creating different policies/laws that protect our environment. • Ecologists study how organisms interact with their environment, and often focus on natural cycles like the water cycle.

Module 1: Well-come to BEAM!

Mentees and mentors will play the Name Game to introduce themselves. Mentees will make name cards for their scientist personas.

Teaching Goals:

- **BEAM:** Five people who show up once a week and teach you cool science lessons.
- **Introductions:** Learn the mentee's names and have the mentees learn yours!

Materials

- Sticky name tags (for mentors)
- Printer paper (mentee nametags)
- Markers

Different Methods for Teaching

1. **Setting the Pace:** For some mentees, this may be their first interaction with BEAM! Ask questions on what they think BEAM stands for, who we are, and what we will be doing throughout the semester. Encourage different answers and ideas!
2. **Shy Mentees:** Mentees may be reluctant to introduce themselves at first! Have mentors go first as an example, and maybe even create your own name tags with the mentees!

Procedure

1. Mentors should wear sticky name tags with their names written on them.
2. Organize mentees into a circle in the classroom
3. Go around the room and have everyone introduce themselves (say their name) and something from some category that starts with the first letter of their name. (For example you could say Jared and Jaguar if your site chose animals).
4. After this, have mentees return to their seats and pass out a piece of construction paper for each mentee, as well as some drawing materials for each table (if applicable).
5. Have mentees make themselves a name tag **with "scientist" before their name at the top** to keep on their desks throughout the semester! (they should put this somewhere safe to save it for the next site) - so we remember everyone!



Figure 1: Nametags!

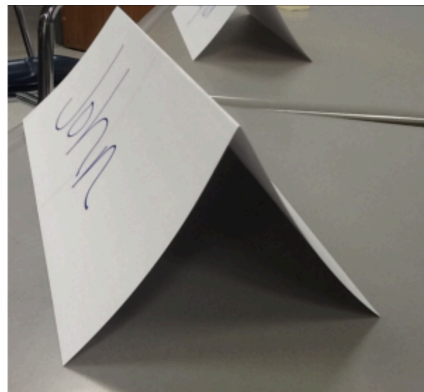


Figure 2: Fold into a tent so you can see their names!

6. Have them draw themselves as a scientist!	
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Classroom Notes

For learning names, try to set up a rule on this first day of site that anytime someone raises their hand, they should say their name. Along the same lines also during activities and builds mentors can walk around and talk to mentees, try to get to know your mentees!

Module 2: It's raining tacos!

In this module, mentees will learn about the water cycle through a group activity simulating a cloud with shaving cream and food dye.

Teaching Goals

1. **Water Cycle:** The movement of water across earth
2. **Evaporation:** Liquid turning to gas
3. **Condensation:** Gas turning into liquid
4. **Clouds:** Small droplets or ice crystals suspended in the sky
5. **Precipitation:** Liquid or solid form of water falling from the sky to the ground
6. **Transpiration:** Movement of water through plants

Materials

- Clear plastic cups
- Water (**fill a water bottle at home just in case!**)
- Shaving cream
- Food dye
- Pipette
- Popsicle stick

Different Methods for Teaching

1. **Utilize the Whiteboard:** If there is a whiteboard available in the classroom, draw a diagram of the process! This will allow mentees to visualize all components of the water cycle
2. **Connecting Activity and Teaching Goals:** Clouds can take in a lot of water, but if they get too heavy, the water comes out or precipitates. The cloud (shaving foam) needs to take in a certain amount of water (dyed water) to make rain.

Procedure

1. In small groups, have each mentee do one of the following steps.
2. In a cup, put some water and ~3 drops of food dye. This will be the “rain” solution.
3. In a separate cup, put water representing the atmosphere.
4. Put a thin layer of shaving cream on top and spread it out with a popsicle stick.
5. With a pipette, slowly put more and more solution until finally it starts raining!
6. Toss the contents down the drain and throw away the cups when finished!



Figure 1: A glass half full



Figure 2: Put some color at the top



Figure 3: Watch it trickle!

Classroom Notes

Have each mentee do a different step, and each one can take a turn pipetting the solution!

Module 3: BYOB (Bring Your Own Bugs)

Mentees will be learning about ecosystems and their living and nonliving components. They will also be introduced to another nutrient cycle: the carbon cycle. Mentees will model ecosystems and nutrient cycles in miniature by making their own terrariums.

Teaching Goals	Materials
1. Ecosystem: a community of living things working together with the non-living things in their environment a. Biotic Factors: the living parts of an ecosystem b. Abiotic Factors: the non-living parts of an ecosystem 2. Carbon: the material that plants, animals, and other living things are made of 3. Carbon Cycle: movement of carbon through the environment a. Photosynthesis: plants breathe in carbon b. Respiration: animals breathe out carbon c. Decomposition: fungi break down carbon from plants and animals	● Mini spray bottles ● Plastic bottle ● Plastic spoon ● Paper funnel ● Pebbles ● Activated charcoal ● Soil ● Chia seeds ● Water ● Paper towels ● Printer paper ● Markers

Different Methods for Teaching

1. **Ecosystem:** An ecosystem is like a classroom! The living components, like the students and teacher, are only able to learn and teach with the help of nonliving components, like pencils, desks, and whiteboards.
2. **Biotic/Abiotic Factors:**
 - a. Have mentees brainstorm different living and non-living things in the environment, like animals, plants, rocks, and water.
 - b. For younger mentees, skip the heavy vocab terms and simply refer to them as living and non-living things.
3. **Carbon Cycle:** Just like the water cycle, carbon moves between different places on Earth. Plants and animals depend on the exchange of carbon between them. This would also be a great diagram to draw out for the mentees.
4. **The Terrarium Connection:** A terrarium is a miniature ecosystem. It is able to stay fully sealed because the nutrients inside, like water and carbon, cycle throughout it.
 - a. Think of a greenhouse - water droplets collect on the walls and “rain down” back onto the plants. It is closed off from the outside, but life can still happen within it!

Procedure

1. Fill plastic spray bottles with water.
2. Each mentee will be making their own terrarium. Pass out one plastic bottle with lid to each mentee.
3. Add 8-10 pebbles to the bottom of the bottle, making a single, flat layer.
4. **Using the funnel**, add a small spoonful of activated charcoal over the rocks to make a thin layer.
5. **With the funnel**, add soil to fill the bottle about halfway.
6. Sprinkle in chia seeds (see Figure 3 for amount). Do not bury them– they need light to germinate.
7. Spray the soil with water until moistened throughout.
8. Securely screw the lid back on the bottle. This is the completed terrarium!
9. Mentees can then draw on the bottle using the provided markers. They can decorate it how they'd like, or draw bugs that they'd like to see in the soil!
10. Keep terrariums in a sunny window/outside for best results! Seeds may germinate within 7-14 days.
11. If time, example of a terrarium that has been alive and kicking since 1972:
<https://www.greenprophet.com/2013/05/terrarium-david-latimer/>



Figure 1: Single layer of pebbles (8-10). Spread charcoal evenly over this layer.



Figure 2: Fill the bottle about halfway with soil.



Figure 3: Sprinkle chia seeds on top of soil layer.



Figure 4: Completed terrarium! You should see some water around the sides, but not a lot of excess.

Classroom Notes

- Put down printer paper/paper towels below the terrariums during assembly for easy spill cleanup.
- For younger sites, mentors should help more with the charcoal and soil.
- This whole process can be done in small groups (1 mentor/group).

Conclusion

- Everyone can be a scientist and BEAM gives a glimpse at what scientists do.
- Terrariums are mini-ecosystems containing various closed cycles, including the water cycle and the carbon cycle!

References

- Module 1 was written by Grace Xie and Grace Kim
- The module 2 activity was inspired by Tina O'Block:
<https://lessons4littleones.com/water-cycle-rain-cycle-science-experiments/>
- Terrariums inspired by PBS:
<https://www.pbs.org/parents/crafts-and-experiments/plant-a-terrarium>

Summary Materials Table

Material	Amount per Site	Expected \$\$	Vendor (or online link)
Printer paper	20-50 sheets	\$0	Amazon
Sticky name tags	1 per mentor	\$0	
Markers		\$0	
Plastic cups	5-6	\$5.51	Amazon
Water			
Food dye	1 tube	\$15.70	Amazon
Shaving cream	1 can	\$31.74	Amazon
Popsicle sticks	5-6	\$0	
Plastic droppers	5-6	\$0	
Plastic bottles	1 per mentee	\$199.96	Amazon
Pebbles	8-10 per mentee	\$99.96	Amazon
Activated charcoal	½ tsp per mentee	\$13.99	Amazon
Soil	⅓ cup per mentee	\$9.97	Amazon
Chia seeds	½ tsp per mentee	\$9.59	Amazon
Paper funnels	5-6	\$9.99	Amazon
Plastic spoons	5-6	\$4.49	Amazon

